

QUANG NGAI, Viet Nam

WMO: 488630

Lat: 15.133N

Lon: 108.783E

Elev: 9

StdP: 101.22

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	18.5	14.3	10.2	19.4	15.3	10.9	19.6	5.0	21.9	4.3	22.2	1.4	320	0.258

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	7.5	36.1	27.6	35.2	27.2	34.5	27.0	28.2	34.0	27.9	33.4	27.6	32.8	1.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	22.4	30.8	26.5	22.1	30.5	26.2	21.6	30.2	91.2	34.5	89.4	33.7	87.9	33.0	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.4	3.8	3.3	16.4	37.2	1.1	1.2	15.7	38.0	15.0	38.7	14.4	39.4	13.6	40.2		
			14.9	28.7	0.9	0.8	14.2	29.2	13.7	29.7	13.2	30.1	12.5	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	22.4	23.3	25.1	27.6	29.6	30.3	29.8	29.5	28.2	26.5	25.2	23.1
	DBStd	3.20	1.86	2.09	2.19	1.68	1.51	1.35	1.33	1.51	1.43	1.25	1.75	2.01
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6110	384	373	469	528	607	608	613	605	547	513	456	407
	CDD18.3	3068	126	140	211	278	348	358	354	347	297	254	206	148
	CDH23.3	32252	412	766	1687	3046	4515	4858	4638	4436	3402	2348	1516	629
Wind	CDH26.7	12947	29	118	429	1137	2167	2516	2257	2092	1300	575	268	59
	WSAvg	1.4	1.4	1.5	1.6	1.5	1.4	1.2	1.2	1.2	1.2	1.3	1.6	1.4
	PrecAvg	2722	111	36	52	35	80	77	78	150	302	685	598	355
	PrecMax	4054	288	111	222	304	236	296	220	547	762	1138	1433	1332
	PrecMin	1829	15	4	4	1	10	15	23	35	63	278	131	35
	PrecStd	658	72	27	50	61	59	68	53	113	175	269	353	316
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.9	30.3	31.7	34.7	36.7	37.3	37.0	36.7	35.0	32.2	31.2	29.6
		MCWB	24.7	25.1	26.0	27.4	27.9	28.2	27.6	27.4	27.7	26.6	26.3	24.8
	2%	DB	27.4	29.2	30.7	33.2	35.4	35.9	35.4	35.4	33.6	31.1	29.8	28.2
		MCWB	23.4	24.4	25.5	26.8	27.4	27.6	27.0	27.3	27.2	26.0	25.4	24.5
	5%	DB	26.4	28.1	29.9	32.2	34.5	35.0	34.4	34.3	32.6	30.2	29.1	27.1
		MCWB	22.9	23.8	25.1	26.3	27.1	27.2	26.8	26.9	26.7	25.6	25.2	23.9
	10%	DB	25.5	27.2	29.1	31.3	33.6	34.1	33.4	33.2	31.7	29.4	28.3	26.2
		MCWB	22.4	23.2	24.7	25.9	26.9	26.9	26.6	26.7	26.4	25.3	24.8	23.4
	0.4%	WB	25.1	25.6	26.7	27.9	28.6	28.8	28.5	28.7	28.2	27.3	26.7	25.6
		MCDB	27.8	29.4	30.5	33.3	34.9	35.5	34.4	34.6	33.4	30.8	29.8	27.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	24.2	24.8	26.1	27.2	28.0	28.1	27.9	28.0	27.6	26.6	26.1	25.0
		MCDB	26.3	28.0	29.7	32.0	34.0	34.1	33.4	33.5	32.4	29.8	28.8	27.1
	5%	WB	23.6	24.2	25.5	26.7	27.6	27.7	27.5	27.5	27.1	26.2	25.7	24.5
		MCDB	25.6	27.1	28.9	31.1	33.0	33.2	32.6	32.6	31.3	29.0	28.1	26.4
	10%	WB	23.0	23.7	25.0	26.3	27.2	27.3	27.1	27.1	26.7	25.8	25.3	23.9
		MCDB	24.8	26.3	28.1	30.2	32.1	32.5	31.9	31.9	30.5	28.4	27.5	25.7
	5% DB	MDBR	4.2	5.4	5.9	6.6	7.4	7.5	7.2	7.0	6.1	4.9	4.2	3.8
		MCDBR	5.9	7.2	7.4	8.1	8.9	8.8	8.5	8.6	7.9	6.5	5.9	5.4
		MCWBR	2.4	3.0	2.9	2.8	2.8	2.7	2.9	2.9	2.9	2.4	2.2	2.3
		MCWBR	5.0	6.7	6.8	7.5	8.3	8.2	7.9	8.1	7.3	5.6	5.1	4.6
Clear-Sky Solar Irradiance	taub		0.472	0.487	0.581	0.545	0.478	0.453	0.438	0.442	0.476	0.522	0.507	0.517
		taud	2.189	2.160	1.925	2.066	2.290	2.374	2.405	2.384	2.289	2.117	2.130	2.044
	Ebn at Noon		819	826	759	785	826	839	853	858	833	785	780	761
		Edh at Noon	142	152	197	171	135	123	119	123	135	157	150	161
All-Sky Solar Radiation	RadAvg		3.44	4.56	5.07	5.82	5.95	5.73	5.45	5.29	4.86	4.00	3.40	2.85
	RadStd		0.43	0.66	0.44	0.39	0.37	0.35	0.46	0.45	0.51	0.45	0.59	0.55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.33	N/A	N/A	+0.60	N/A	N/A	N/A	+228	+226

Nomenclature: See separate page